# AI PM Learning Roadmap

v 3.0, October 2025

## 1. Basic Concepts

### Machine Learning:
- ☐ Supervised Learning
- ☐ Unsupervised Learning
- ☐ Reinforcement Learning
- ☐ Deep Learning

### Free resources:
- WTF is AI Product Manager
- LLM Visualization

### Architectures:
Neural Networks > Transformers > LLMs

### Models:
LLM, LCM, LAM, MoE, VLM, SLM, MLM, SAM

## 2. Prompt Engineering

### Free guides & courses:
- GPT-4.1, GPT-5 Prompting Guides
- Anthropic Prompt Eng.
- Prompt Eng. by Google
- System prompt, Claude 4
- Anthropic Prompt Generator
- Anthropic Prompt Library
- Anthropic Prompting Course

### Techniques:
CoT, Roles, Examples, Step-by-Step, Constraints, Chain prompts, Autonomy, Reflect, XML #, Persist

## 3. Fine-Tuning

### Methods:
- ☐ Supervised (SFT)
- ☐ Preference-based (DPO, ORPO, KTO)

### Best tools:
- OpenAI Platform
- Hugging Face AutoTrain
- LLaMA-Factory

### Terms:
- ☐ Training data
- ☐ Validation data
- ☐ Epoch
- ☐ Batch size
- ☐ Learning rate
- ☐ Beta (for DPO)

| Key Metric | What it Tells You |
|---|---|
| Training Loss ↓ | The model is learning from data. |
| Training Accuracy ↑ | Better at predicting tokens it has seen. |
| Validation Loss ↓ | It's generalizing well to new data. |
| Validation Accuracy ↑ | It's predicting new tokens well. |

## 4. RAG (Retrieval-Augmented Gen.)

### Vector DBs
- Weaviate
- Pinecone
- + pgvector

### Document DBs
- OpenSearch
- elastic
- MongoDB.

### Knowledge graphs
- Weaviate
- neo4j

### Vector DB vs. Document DB

Use **vector store** to store embeddings and perform fast semantic search over chunks. Use **document DB** to store the full content and metadata (e.g., document url, author, date).

## 5. AI Agents & Agentic Workflows

n8n, zapier, crewai, Cassidy, Flowise, make, LangChain, Lindy, LAMINI, LlamaIndex, LangGraph, haystack by deepset, AutoGen

### Techniques:
Tool use, MCP, A2A, RAG, Agent Architectures

### Free reports:
- Google Agent Companion
- Anthropic Building Agents
- IBM Agentic Process Automation

## 6. AI Prototyping & Building

### No-code first:
Lovable, Bolt, Databutton, Firebase, Base44

### IDE-first:
Replit, v0, Windsurf, Cursor, Claude code

### AI copilots / IDE extensions:
GitHub Copilot, Codex, Jules

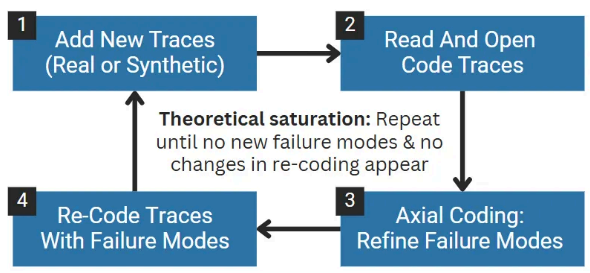
### Infrastructure Tools
supabase, Firebase, clerk, GitHub, OpenRouter, PromptLayer, docker, DigitalOcean, netlify, arize, truera, Streamlit, VAPI, ElevenLabs

## 7. Foundational Models

Claude, DeepSeek, Llama, Grok, ChatGPT, Qwen3, Mistral, Gemini

## 8. AI Evaluation Systems

### Error Analysis:




### Techniques:
Unit Tests, LLM Judge, Error Analysis, TNR, TPR, Human Eval, Model Eval

- Mastering AI Evals: A Free, Complete Guide for PMs
- AI Evals: How to Find The Right AI Product Metrics
- The Guide to AI Observability and Evaluation Platforms

## 9. Other Resources

- The AI Strategic Lens Framework
- Building AI Product Strategy
- AI Product Distribution Framework
- How to Build an AI Product Team
- AI PRD: A free template
- Generative AI Guide
- ChatLLM (abacus.ai):
- MCP.so: Huge collection
- Anthropic MCP Servers
- microsoft/markitdown